

University of Exeter

MSc Data Science

MSc Research Project Proposal

LLM Hallucination Detection and Verification

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Abstract

Large language models can produce fluent answers that are nevertheless factually incorrect, while asking the same model to check its own response may reproduce the same underlying error. This project investigates whether using another LLM as a verifier provides a more reliable signal, and whether any benefit depends on the relationship between the generator and verifier. Using SimpleQA Verified, models from several LLM families will generate factual answers that are graded automatically against benchmark reference information. The same answers will then be assessed through self-verification, within-family verification and cross-family verification under a common protocol. Detection performance will be compared across generator–verifier pairings, while overlap in the models’ factual errors will be examined to see whether shared failures are associated with weaker verification. The study aims to clarify when a second model adds useful evidence and when agreement between models may instead reflect errors they have in common.

Generative AI declaration

I acknowledge the following uses of GenAI tools in this assessment:

- ☒ I have used GenAI tools to:
- ☒ develop ideas.
- ☒ assist with research or gathering information.
- ☒ help me understand key theories and concepts.
- ☐ identify trends and themes as part of my data analysis.
- ☒ suggest a plan or structure for my assessment.
- ☐ give me feedback on a draft.
- ☐ generate images, figures or diagrams.
- ☒ proofread and correct grammar or spelling errors.
- ☐ generate citations or references.
- ☒ Other: help with checking and debugging the Python code used in the experimental pipeline.
- ☐ I have not used any GenAI tools in preparing this assessment.

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1 Introduction

Large language models (LLMs) are now used for tasks ranging from question answering and information retrieval to writing and decision support. Their fluency can make factual errors particularly difficult to notice: an answer may read naturally and confidently while still containing unsupported or incorrect claims. These outputs are commonly described as hallucinations and remain an important obstacle to reliable LLM use, especially when users cannot easily check the information themselves [1]. Factual errors are not always random either. TruthfulQA, for example, showed that language models can reproduce common misconceptions and false beliefs found in human-generated text [2].

A practical response is to check an answer after it has been generated, either with the same model or with another LLM. This is attractive in a black-box setting because it does not require access to internal model parameters. SelfCheckGPT takes this approach by comparing several samples from the same model and treating disagreement as a signal of possible hallucination [3]. LLMs have also been used more directly as judges of model outputs, although such judgements are not neutral. Position and verbosity biases have been observed, along with cases where a judge appears to favour outputs associated with itself [4]. This makes self-verification useful, but also raises the question of how much confidence should be placed in a model checking its own answer.

Work on self-correction points to the same concern. Huang et al. found that asking an LLM to reconsider its reasoning without external feedback did not reliably improve performance and could sometimes turn a correct response into an incorrect one [5]. Other approaches introduce a more separate source of verification. Chain-of-Verification, for instance, generates verification questions independently of the original response before revising it [6]. More recent work has used a second model directly. Xue et al. found that cross-model consistency can provide information beyond self-consistency [7], while FINCH-ZK uses responses from different model families to identify fine-grained factual inconsistencies [8].

Using another model, however, does not automatically make the judgement independent. LLMs can share training sources, architectural relationships and broader patterns of learned behaviour. In a study of more than 350 models, Kim et al. found substantial correlation between model errors, including among capable models from different architectures or providers [9]. A verifier that shares the generator’s misconception may therefore accept the same incorrect answer. This makes the choice of generator–verifier pairing important, and also raises a more specific question: whether within-family and cross-family verification behave differently when the models share factual failure patterns.

This project will investigate the reliability of self- and cross-model verification for factual LLM outputs, focusing on generator–verifier pairing, model-family relationships and shared factual errors. It does not assume that using another model, or a model from another family, will necessarily improve detection. Instead, the aim is to examine when an external verifier adds useful information and when the models fail together. The main research question is:

To what extent does cross-model verification improve hallucination detection over self-verification, and how does this vary across within-family and cross-family generator–verifier pairings and with shared factual errors?

The study will use a reproducible black-box setting in which LLMs generate answers to factual questions and the same answers are then checked through self-, within-family and cross-family verification. Verification decisions will be evaluated against independently established ground truth, while the models’ factual error patterns will be compared separately. In this way, the project can examine not only whether cross-model verification helps, but also what makes one verifier more useful than another.

2 Background and Related Work

2.1 Hallucination Detection in Factual Question Answering

Hallucination broadly refers to generated content that is unsupported by available evidence or conflicts with established facts [1]. This project focuses specifically on factual question answering, where a generated answer can be checked against a known reference or other reliable evidence. This narrower setting allows verification methods to be evaluated against external ground truth rather than relying only on another model’s judgement.

Black-box hallucination detection is particularly relevant because many deployed LLMs do not expose internal weights, hidden states or token probabilities. SelfCheckGPT addresses this by sampling several responses from the same model and measuring whether factual claims remain consistent across generations [3]. The underlying idea is that reliable knowledge should be relatively stable, while hallucinated claims are more likely to vary.

This approach is useful, but consistency does not necessarily imply correctness. Repeated generations come from the same model and may reproduce the same misconception. Self-consistency can therefore provide an uncertainty signal without guaranteeing independent evidence of factual truth.

2.2 From Self-Refinement to Self-Verification

The broader idea of asking an LLM to examine its own output has been explored in several forms. Self-Refine uses the same model to generate an answer, critique it and revise it iteratively, and reports improvements across a range of tasks without additional training [10]. This suggests that self-feedback can be useful when the model is able to identify weaknesses in its initial response.

The benefit is not universal. Huang et al. found that intrinsic self-correction in reasoning tasks did not reliably improve performance and could sometimes change an initially correct answer into an incorrect one [5]. Related concerns appear in LLM-based judging: Zheng et al. showed that model judgements can be affected by systematic biases, including response position and verbosity [4]. These findings suggest that a model’s ability to produce an answer does not automatically make it a neutral or reliable verifier of that answer.

Other methods introduce information that is more independent from the original generation. Chain-of-Verification separates the initial response from later verification questions before revising the answer [6]. CRITIC allows the model to consult external tools such as search engines or code interpreters during the critique process [11]. Together, these approaches indicate that verification reliability may depend not only on the act of checking an answer, but also on where the verification signal comes from.

2.3 Cross-Model Verification

Cross-model verification replaces or supplements self-checking with a second LLM. The generator first produces an answer, while another model assesses whether that answer should be trusted. This can potentially introduce information that is unavailable when the generator only re-examines its own output.

Xue et al. compared self-consistency and cross-model consistency in black-box hallucination detection and found that an additional verifier can provide complementary information beyond repeated sampling from a single model [7]. Their two-stage procedure also showed that external verification can be applied selectively to uncertain cases rather than to every response.

FINCH-ZK similarly uses outputs from different model architectures and semantically equivalent prompts to identify fine-grained factual inconsistencies [8]. Its results support cross-model consistency as a practical hallucination-detection strategy. These studies establish that a second model can be useful, but they leave open a more specific question: how much that benefit depends on the relationship between the generator and verifier.

A verifier may help because it is more capable, because it has a different error profile, or both. Models from different families may provide more diverse evidence in some cases, but simply choosing a different family does not guarantee an independent judgement. This makes both the individual pairing and the relationship between model families relevant to evaluation.

2.4 Correlated Errors and Generator–Verifier Pairing

The assumption of independence is important when agreement between models is treated as evidence of correctness. LLMs may share training sources, architectural families, optimisation practices or providers, and different systems can therefore make similar mistakes.

Kim et al. analysed more than 350 LLMs and found substantial correlation in model errors [9]. On one evaluated leaderboard dataset, when two models were both wrong, they produced the same error in approximately 60% of those cases. Shared architecture and provider were associated with stronger correlation, although substantial overlap could remain even across otherwise different models.

This has a direct consequence for hallucination verification. If both generator and verifier share the same factual misconception, the verifier may endorse an incorrect answer rather than expose it. A different pairing may provide more complementary evidence if the models’ failure patterns are less strongly aligned. Comparing models within the same family and across different families offers one way to examine this relationship, while the actual overlap in their factual errors provides a more direct measure of how similar their failures are.

Existing work therefore points to two findings that need to be considered together: cross-model verification can improve hallucination detection [7, 8], while model errors themselves can be correlated [9]. The present project builds on this tension by examining how verification performance changes across within-family and cross-family generator–verifier pairings, and whether shared factual errors help explain those differences.

3 Research Gap, Aim and Research Questions

3.1 Research Gap

Recent studies show that cross-model signals can improve hallucination detection beyond repeated sampling from a single model [7, 8]. However, using a second model does not guarantee an independent judgement. Errors across LLMs can be correlated, particularly when models share related architectures, providers or other aspects of their development [9]. If a generator and verifier tend to fail on the same factual questions, agreement between them may add little evidence of correctness.

Existing work has therefore established two relevant findings: cross-model verification can provide useful complementary information, while model errors themselves may overlap. What remains less clear is how strongly verification performance depends on the relationship between the generator and verifier. In particular, models from the same family may behave differently from models drawn from different families, but family membership alone does not guarantee either similar or independent errors. Comparing within-family and cross-family pairings therefore provides a more structured way to examine when an additional verifier is genuinely useful.

3.2 Aim and Research Questions

The aim of this project is to evaluate the reliability of self- and cross-model verification for factual LLM outputs, with particular attention to model-family relationships, generator–verifier pairing and shared factual errors.

The study will address three questions:

RQ1: To what extent does cross-model verification improve hallucination detection over self-verification in LLM-generated factual answers?

RQ2: How does hallucination-detection performance vary between within-family and cross-family generator–verifier pairings?

RQ3: How is cross-model verification performance related to the overlap in factual errors between generator and verifier models?

These questions do not assume that cross-model or cross-family verification will always perform better. Cases in which another model provides little benefit, or introduces additional errors, are equally relevant to understanding when external verification is reliable.

3.3 Objectives and Expected Contribution

The project will establish a reproducible factual question-answering setting in which the same generated answers are assessed through self-verification and verification by other models. Models from more than one family will be included so that within-family and cross-family generator–verifier pairings can be compared using independently established ground truth.

Performance will be examined across these pairings together with the models’ factual error profiles. Error-level analysis will consider cases detected only by an external verifier, errors missed by both models, and correct answers incorrectly rejected. The overlap between models’ factual errors will then be used to examine whether weaker or stronger verification performance is associated with shared failure patterns.

The expected contribution is a controlled empirical account of when a second model provides useful complementary evidence and whether this depends on the relationship between the models being paired. Any practical conclusion about verifier selection will be based on consistent behaviour across multiple pairings rather than a single model comparison.

4 Methodology and Experimental Design

4.1 Experimental Overview

The project will use a controlled black-box setting in which LLMs take separate roles as generators and verifiers. A generator will first answer factual questions from SimpleQA Verified without external search or retrieval. The resulting answers will then follow two parallel paths: they will be graded against the benchmark reference information to establish ground truth, and they will be passed to a set of verifiers for self-, within-family and cross-family checking.

The overall workflow is shown in Figure 1.

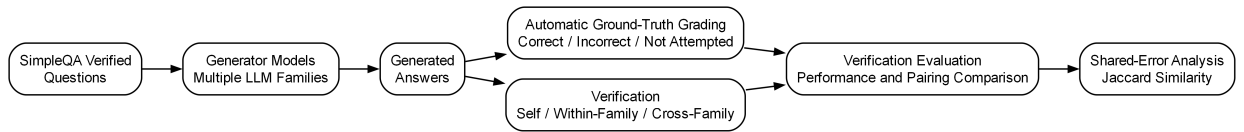


Figure 1: Overview of the experimental pipeline, covering answer generation, ground-truth grading, model verification, performance comparison and shared-error analysis.

The same generated answer will be used across the different verification settings so that comparisons are made on a common basis. Each verifier will receive the original question and candidate answer, but not the benchmark reference answer. Self-verification will also be carried out in a fresh context, preventing the model from retaining the history of its original generation.

4.2 Benchmark and Ground-Truth Labelling

SimpleQA Verified will be used as the main factual question-answering benchmark [12]. It contains 1,000 fact-seeking questions with reviewed reference answers and is derived from the larger SimpleQA

benchmark [13]. Its short-answer format is useful here because the generated responses can be judged against external factual references without requiring long-form subjective assessment.

Each selected generator will answer the same benchmark questions under fixed generation settings. Responses will be assigned one of three labels: *correct*, *incorrect* or *not attempted*. An incorrect attempted answer will form a positive hallucination case for the later detection task, while an explicit refusal or statement of uncertainty will be treated separately as an abstention.

For the final experiment, ground-truth grading will be designed to run automatically at scale rather than relying on manual case-by-case checking. The grading procedure will use the benchmark reference information as its basis and will remain separate from the experimental verifiers. The exact automatic grading method will be fixed after development testing, particularly for answers that are factually equivalent to the reference but differ in wording. The reference answer itself will never be included in the verification prompt.

4.3 Model Families and Generator–Verifier Pairings

The final experiment will include models from several LLM families rather than relying on a single generator–verifier pair. The current plan is to use three families, subject to model availability and stable access through the chosen inference platform. Where suitable models are available, more than one model from a family will be included so that within-family verification can be compared with verification across families.

For each primary generator, the experiment will include its own self-verification, at least one within-family verifier where possible, and verifiers drawn from the other selected families. Pairings will be treated as ordered: model A verifying model B is not assumed to be equivalent to model B verifying model A. An exhaustive all-by-all comparison is not required. Instead, a fixed set of pairings will be selected before the main experiment so that both within-family and cross-family verification are represented without making the experiment unnecessarily large.

Model capability is a possible confounding factor because a stronger verifier may perform better regardless of its family. The selected models will therefore be kept as comparable as practical, and each model’s independent factual performance on the benchmark will also be reported. Family membership will be treated as one feature of a pairing rather than as an explanation by itself. Exact model identifiers, versions and inference settings will be fixed before the final evaluation.

4.4 Verification Protocol

Every verifier will use the same verification prompt and output format. Given a factual question and a candidate answer, it will decide whether the answer should be accepted as factually correct, returning a binary label such as **CORRECT** or **INCORRECT**. Generation and verification settings will be kept fixed where the platform permits this. Invalid or unparseable outputs will be recorded rather than silently removed.

Development runs will use a smaller sample before the full benchmark is processed. This allows prompts, parsing, model access and the automatic grading procedure to be checked without repeatedly running the complete experiment. The existing Gemma–GPT-OSS pilot serves as an early version of this process and will remain separate from the final evaluation.

Once the model set, prompts and grading procedure have been fixed, the main experiment will be run at larger scale. The intention is to use the full SimpleQA Verified set for the final evaluation, subject to practical model-access and computation constraints.

4.5 Shared-Error Analysis and Reproducibility

Models used in the pairing analysis will also answer the benchmark independently so that a factual error profile can be constructed for each one. For a given pair, the sets of questions answered incorrectly by the two models will be compared and related to the verifier’s ability to identify the generator’s

hallucinations. This makes it possible to examine whether models that fail on many of the same questions also provide less useful verification of one another.

Family membership will not be treated as a substitute for this error-level comparison. Models from the same family may behave differently, while models from different families may still share similar factual mistakes. The analysis will therefore consider both the family relationship and the measured overlap in factual errors.

Questions, generated answers, ground-truth labels and verifier decisions will be stored in structured files together with model identifiers, settings and run information. The generation, grading, verification and evaluation scripts will remain version-controlled in the project repository. Intermediate results will be saved incrementally so that interrupted runs can be resumed without repeating completed work.

5 Evaluation and Success Criteria

Verification performance. The factual performance of each generator will first be reported as the proportions of *correct*, *incorrect* and *not attempted* responses. For hallucination detection, only attempted answers will enter the main binary evaluation: incorrect answers will be treated as positive cases and correct answers as negative cases. Accuracy, precision, recall, F1-score and false-positive rate will be reported, while abstentions and unparseable verifier outputs will be kept separate.

The class distribution will be reported alongside these metrics. This is important because the preliminary 20-question pilot contained incorrect answers and abstentions but no correct attempted answers, making some measures, especially precision and false-positive behaviour, uninformative. The final experiment will therefore use a larger evaluation set, and all performance measures will be interpreted in the context of the observed class balance.

Pairing and error-level analysis. Self-, within-family and cross-family verification will be compared on the same generated answers wherever possible. For each generator, the performance of its own self-verification will provide a baseline against which the selected external verifiers can be compared. Results will be reported both for individual ordered generator–verifier pairings and, where appropriate, grouped by within-family and cross-family verification.

Case-level disagreements will also be examined. For any two verification methods applied to the same answers, cases can be separated into those where both are correct, only one is correct, or both are wrong. Where the number of comparable cases is sufficient, McNemar’s test will be used for paired differences in verification accuracy, together with confidence intervals for the main performance measures. Family-level summaries will be interpreted alongside the individual pairings so that differences are not attributed to model family alone when verifier capability may also contribute.

For RQ3, each model’s independently generated factual errors will be represented as a set of incorrectly answered benchmark questions. Jaccard similarity will be used to measure the overlap between the error sets of each generator–verifier pair. These overlap values will then be considered alongside the corresponding verification results to examine whether stronger shared-error patterns are associated with weaker complementary verification.

Success criteria. Project success will not depend on cross-model or cross-family verification outperforming self-verification. The study will be considered successful if it provides a reproducible comparison of self-, within-family and cross-family verification across multiple generator–verifier pairings using independently established ground truth, and if the analysis can identify where their behaviour differs and how those differences relate to shared factual errors.

A stronger outcome would be the identification of consistent pairing or family-level patterns that could inform verifier selection in black-box systems. Any such conclusion will only be drawn if it is supported across more than one pairing and is not driven by a small number of individual cases.

6 Risks, Ethics and Limitations

Risks and mitigation. The main practical risk is dependence on remotely hosted black-box models. Cloud availability, usage limits and model versions may change during the project, and individual requests can occasionally fail or take unusually long to complete. Exact model identifiers, dates and generation settings will therefore be recorded for each experiment. Requests will use timeouts and limited retries, while intermediate results will be saved incrementally so that interrupted runs can be resumed rather than started again.

Model availability also affects the planned family-level comparison. It may not be possible to obtain equally suitable models from every family, particularly for within-family pairings. The final model set will therefore be fixed before the main evaluation, after small development runs have confirmed that the selected models are accessible and can follow the required response format.

A second risk concerns the quality and balance of the evaluation data. Benchmark errors or ambiguous reference answers could lead to incorrect ground-truth labels. An evaluation set with very few correct or incorrect generated responses would also make some verification metrics difficult to interpret, as already seen in the small pilot. Reference-based grading will therefore remain separate from the experimental verifiers, and the grading procedure will be tested during development before it is applied at scale. The distribution of correct, incorrect and abstained responses will also be reported before the main results are interpreted.

Ethical considerations. No human participants will be recruited and no new personal or sensitive data will be collected. The study will use publicly released benchmark questions and model-generated responses, subject to the relevant dataset licences and service terms. Generated text may occasionally contain inaccurate, biased or otherwise undesirable material. Such outputs will only be retained where they are needed for analysis and will not be presented as factual information.

Experiment files will be stored within the project workspace. Only code, configuration information and material needed for reproducibility will be committed to the repository.

Limitations. The study is limited to factual question answering and a relatively small number of black-box LLM families. Its findings may therefore not generalise to long-form generation, retrieval-augmented systems or models with substantially different capabilities.

The family comparison also needs to be interpreted with care. Models within the same family can differ in size, training and capability, while models from different families may still share data or similar failure patterns. A within-family or cross-family label is therefore only one feature of a pairing, not an explanation by itself. Pairing results will be considered alongside each model’s independent factual performance and the measured overlap in their errors.

Finally, hosted models can change over time. The experiments will therefore provide a reproducible record of the tested configurations, rather than a claim that exactly the same outputs will remain obtainable indefinitely.

7 Project Plan and Deliverables

The remaining work will move from finalising the experimental design to full model runs, analysis and dissertation writing. The next step is to confirm the model families and generator–verifier pairings, together with the benchmark, prompts and grading procedure. Small development runs will be used before the main experiment so that access or formatting problems can be found early.

Stage	Main work	Output
1	Finalise model families, pairings, prompts and evaluation procedure	Fixed experiment protocol
2	Generate benchmark answers and establish independent ground-truth labels	Generation dataset and labels
3	Run self-, within-family and cross-family verification; analyse pairing and shared-error effects	Evaluation results and case analysis
4	Interpret results, complete the dissertation and reproduce key analyses	Final dissertation and reproducible project repository

The main deliverables will be the dissertation, a version-controlled Python implementation of the experimental pipeline, structured experimental results, and sufficient configuration information to reproduce the principal analyses. The existing pilot will remain as evidence that the pipeline is workable, but it will be kept separate from the final evaluation.

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